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10 December 1959
SUPERSEDING
FA(FC)-PD-272
24 August 1959

MILITARY SPECIFICATION

PERISCOPES XM32 AND XM35; PACKAGING OF

1. SCOPE

1.1 This specification covers the packaging and packing of Periscopes XM32 and XM35 for each level of protection.

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on date of invitation for bids, form a part of this specification:

SPECIFICATIONS

Federal

O-E-760	Ethyl Alcohol (Ethanol)
UU-P-313	Paper, Lens Tissue
UU-T-106	Tape, Masking; Paper (Pressure-Sensitive)
UU-T-111	Tape, Paper Gummed (Sealing and Securing)
CCC-C-440	Cloth, Cotton, Cheesecloth, Bleached and Unbleached
JJJ-C-561	Cotton; Sterile and Nonsterile
PPP-B-585	Boxes; Wood, Wirebound
PPP-B-601	Boxes, Wood, Cleated Plywood
PPP-B-621	Boxes, Wood, Nailed and Lock-Corner
PPP-B-636	Boxes, Fiber
PPP-P-291	Paperboard, Wrapping, Cushioning
PPP-T-76	Tape, Pressure-Sensitive Adhesive, Paper, Water Resistant

FSC 1240

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Military

MIL-P-116	Preservation, Methods of
MIL-B-117	Bags, Interior Packaging
MIL-B-121	Barrier Material, Greaseproofed Flexible (Waterproofed)
MIL-B-130	Barrier Material, Paper, Noncorrosive
MIL-L-6085	Lubricating Oil; Aircraft Instrument, Low Volatility
MIL-C-7769	Cushioning Material, Bound
MIL-P-13988	Paper, Lens Tissue, Antitarnish, Wrapping

STANDARDS

Military

MIL-STD-105	Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-129	Marking for Shipment and Storage

(Copies of specifications and standards required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Materials and methods. - The materials, methods and procedures used in the packaging of items shall be in accordance with this specification. Materials and methods not covered by a specification shall be suited to the intended use and subject to the approval of the contracting officer.

3.2 Size of containers. - Bags, interior containers and exterior containers shall be of a minimum size that will adequately contain the items.

3.3 Preservation and packaging.

3.3.1 Level A.

3.3.1.1 Cleaning.

3.3.1.1.1 Optics.

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3.3.1.1.1.1 The following materials and equipment shall be used in the cleaning of the exposed glass surfaces of the optics:

Materials and equipment	Specification
Alcohol, ethyl	Grade II of Specification O-E-760
Bottle, glass, 4 ounce	----------
Brush, camel's hair	----------
Cheesecloth	Specification CCC-C-440
Cloth, batiste	----------
Cotton, absorbent	Grade A, class 1 of Specification JJJ-C-561
Paper, lens tissue	Specification UU-P-313
Sticks, orange or equal	----------
Syringe, rubber	----------

3.3.1.1.1.2 Cleaning the exposed glass surfaces shall be accomplished in the following manner:

(a) Loose particles of dust shall be removed from the surface of the optic with a camel's hair brush.

(b) The surface to be cleaned shall be wiped with a circular motion using cotton, lens paper or freshly laundered cheesecloth which is saturated with ethyl alcohol. The surface shall be dried with clean cotton, lens paper or freshly laundered cheesecloth.

(c) A swab shall be made by wrapping one piece of lens tissue paper around the end of an orange stick or equal. One or two drops of alcohol shall be added on the tip of the swab. The swabbing operation shall begin at the center of the polished surface. A light, downward pressure shall be exerted on the swab and with a circular motion gradually increase the radius of area being cleaned until the entire surface has been swabbed. This operation shall be repeated until no trace of dirt, lint or smears are perceptible.

(d) A rubber syringe shall be used as a bellows to remove any residual dust or lint that may remain on the cleaned surface.

(e) Immediately after cleaning, the optics shall be covered or wrapped with lens tissue paper conforming to Specification MIL-P-13988. The lens tissue paper shall be secured in place with pressure-sensitive tape conforming to Specification UU-T-106.

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3.3.1.1.2 Other surfaces. - Cleaning Process C-1 of Specification MIL-P-116 shall be followed to clean all other exterior surfaces of the periscope.

3.3.1.2 Drying. - Immediately after cleaning, the item shall be dried thoroughly in accordance with any applicable procedure specified in Specification MIL-P-116 that will not damage the item.

3.3.1.3 Preservative application. - The exposed metal surfaces of the coupling wedge shall be coated with preservative conforming to Specification MIL-L-6085 (P-17 of Specification MIL-P-116). The preservative shall be applied by the brushing procedure specified in Specification MIL-P-116. All excess preservative shall be removed.

3.3.1.4 Unit packaging. - Each periscope shall be packaged as specified herein, which is in conformance with Method IA-8 of Specification MIL-P-116.

3.3.1.4.1 Wrapping preservative coated surfaces. - The preservative coated surfaces of the wedge shall be wrapped with greaseproofed barrier material conforming to type II, grade A, class 2 of Specification MIL-B-121. The barrier material shall be secured in place with pressure-sensitive tape conforming to Specification UU-T-106.

3.3.1.4.2 Wrapping the periscope. - The periscope shall be wrapped with barrier material conforming to type I of Specification MIL-B-130 and over-wrapped with paperboard cushioning material conforming to type I, class 1 of Specification PPP-P-291. The wrap shall be secured in place with gummed tape conforming to class 2 of Specification UU-T-111.

3.3.1.4.3 Water-vaporproof bag. - The wrapped periscope shall be placed in a bag conforming to type 1, class e of Specification MIL-B-117. The bag shall be heat sealed in such a manner as to pass the heat seal and quick leak tests of Specification MIL-P-116. Air shall be exhausted from the bag prior to final closure.

3.3.1.4.4 Fiberboard box. - Each bagged periscope shall be cushioned and immobilized (specified in 3.3.1.4.5) in a fiberboard box conforming to class 2 or 3, compliance symbol V2s, V3s or V3c and any style, except FTC, TS, OPF and FPF of Specification PPP-B-636.

3.3.1.4.5 Cushioning, immobilizing and closure of fiberboard box. - The interior surfaces of the fiberboard box (sides, ends, top and bottom) shall be lined with 4. inch thick bound cushioning material conforming to type IV of Specification MIL-C-7769. The bagged periscope shall fit snugly in the bound cushioning material. The fiberboard box shall be closed with pressure-sensitive tape

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conforming to Specification PPP-T-76 in accordance with appendix of Specification PPP-B-636.

3.3.2 Level C.

3.3.2.1 Cleaning and drying. - Cleaning and drying shall be as specified in 3.3.1.1 and 3.3.1.2.

3.3.2.2 Unit packaging. - Each periscope shall be packaged as specified in 3.3.1.4.1 to 3.3.1.4.5 inclusive, except as follows, which is in conformance with Method III of Specification MIL-P-116:

(a) Since preservative is not required, the greaseproofed barrier material, specified in 3.3.1.4.1, shall be omitted.

(b) The water-vaporproof bag, specified in 3.3.1.4.3, shall be omitted.

(c) A fiberboard box conforming to type I or II, class 1, any style, except FTC, TS, OPF and FPF, of Specification PPP-B-636 shall be used in lieu of the fiberboard box specified in 3.3.1.4.4. The box shall be closed as specified in the appendix to the box specification.

3.4 Packing.

3.4.1 Level A. - Two unit packaged periscopes shall be immobilized in a shipping container conforming to one of the following specifications:

PPP-B-601 - Style A or B (oversea type) - type II or III, class 2 plywood
PPP-B-621 - Class 2 (oversea), style 2 or 4

3.4.2 Level B. - Two unit packaged periscopes shall be immobilized in a shipping container conforming to one of the following specifications:

PPP-B-585 - Any style, except style 1, class 1, construct for type 3 load
PPP-B-601 - Any style under domestic type
PPP-B-621 - Class 1 (domestic), style 2 or 4

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3.4.3 Level C. - A number of unit packaged periscopes shall be immobilized in a shipping container conforming to one of the following specifications:

PPP-B-585 - Any style except style 1, class 1, construct for type 3 load
PPP-B-601 - Any style under domestic type
PPP-B-621 - Class 1 (domestic), any style
PPP-B-636 - Type I or II, class 1, any style, except FTC, TS, OPF and FPF

3.4.3.1 The gross weight for a Level C wood shipping container shall not exceed 200. pounds. The gross weight for a fiberboard shipping container shall not exceed the weight limits specified in Specification PPP-B-636.

3.4.4 Strapping. - Shipping containers shall be strapped as specified in the applicable container specification.

3.5 Marking.

3.5.1 Interior packages. - Each bag and each box shall be marked in accordance with Standard MIL-STD-129. The serial number shall be added in the identification data on each Level A unit package and omitted on the Level C unit package. Fragile markings shall be added on each fiberboard container.

3.5.2 Shipping containers. - Each shipping container shall be marked in accordance with Standard MIL-STD-129. The serial numbers of the periscopes in a Level A or B shipping container shall be indicated in the identification data. Fragile labels shall be affixed to the container in accordance with Standard MIL-STD-129.

3.5.3 Stock number. - The stock number to be added in the identification data shall be obtained from the contracting officer.

3.6 Preproduction pack. - Prior to quantity production, a preproduction pack, consisting of a complete and packed shipping container, shall be inspected to determine conformance with the applicable requirements of this specification. A rough handling test is required.

3.7 Workmanship. - All operations involved in accomplishing corrosion prevention, preservation, packaging, and packing by the procedures specified herein shall be in accordance with the highest grades of practice associated with this type of work.

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4. QUALITY ASSURANCE PROVISIONS

4.1 Contractor inspection. - Unless otherwise specified herein, the supplier is responsible for the performance of all inspection requirements prior to submission for Government inspection and acceptance. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. Inspection records of the examinations and tests shall be kept complete and available to the Government as specified in the contract or order.

4.2 Contractor's quality assurance system. - The contractor shall provide and maintain an effective quality assurance system acceptable to the Government covering the supplies under the contract. A current written description of the system shall be submitted to the contracting officer prior to initiation of production. The written description will be considered acceptable when it provides:

(a) The quality assurance required by the applicable drawings and specifications listed in the contract or order.

(b) Contractor's quality standards of acceptability.

(c) Procedures for continually evaluating product quality.

(d) Procedure for continually reporting evidence of actual product quality.

(e) Measures for preventing any significant departure of product quality from standards.

4.2.1 The contractor will not be restricted to the inspection station or to the method of inspection listed provided that an equivalent control is included in the approved quality assurance procedure. In cases of dispute as to whether or not certain procedures of the system provide equal assurance, the comparable procedure of the detail specification shall be used. The contractor shall notify the Government of, and obtain approval for, any change to the written procedure that might affect the degree of assurance required by the detail specification or other applicable documents referenced herein.

4.3 Government verification. - All quality assurance operations performed by the contractor will be subject to Government verification at unscheduled

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intervals. Verification will consist of (a) surveillance of the operations to determine that the practices, methods, and procedures of the written inspection plan are being properly applied, and (b) Government product inspection to measure quality of product offered for acceptance. Deviation from the prescribed or agreed upon procedures, or instances of poor practices which might have an affect upon the quality of the product, will be immediately called to the attention of the contractor. Failure of the contractor to promptly correct deficiencies discovered shall be cause for suspension of acceptance until correction has been made or until conformance of product to prescribed criteria has been demonstrated. To avoid interference with operations, the contractor shall designate a responsible official or officials, to whom the Government inspector will report such instances.

4.4 Inspection provisions. - Quality assurance provisions specified in Specification MIL-P-116 shall apply where applicable and are not in conflict with the provisions specified herein.

4.4.1 Control of materials. - Materials shall be tested to determine conformance to contractually applicable specifications. Such tests may be conducted in the prime contractor's or subcontractor's facilities, or in other suitably equipped laboratories. The test results shall be available to the Government inspector for review. Certified test reports, identifiable with the material, may be accepted in lieu of repeated tests. The Government reserves the right, in any case, to require retests to assure that the materials are acceptable.

4.4.2 Lot formation. - Unless otherwise specified by the contracting officer, or his representative, inspection lot sizes and lot formation shall be in accordance with Standard MIL-STD-105.

4.4.3 Examination. - The classification of defects in table I and II shall constitute the minimum inspection requirements to be performed by the contractor prior to acceptance or rejection by lot. The Government reserves the right to inspect for any applicable requirement, and to reject individual nonconforming items.

4.4.3.1 Rejected lots shall be screened for all defective characteristics. After corrections have been made and the Government inspector informed of the corrective action taken, the lot shall be resubmitted for acceptance as outlined in Standard MIL-STD-105.

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TABLE I - CLASSIFICATION OF DEFECTS

For Normal Inspection use Level II, and Sampling Plan Table IV-A of Standard MIL-STD-105

For Reduced Inspection use Level I, and Sampling Plan Table IV-A of Standard MIL-STD-105

AQL values are expressed in percent defective

MAJOR: AQL 1.5		Req't Par.	Test Method
1.	Improper cleaning, optical-mechanical surfaces (Level A, C)	3.3.1.1 3.3.2.1	Visual MIL-P-116
2.	Improper drying mechanical surfaces (Level A, C)	3.3.1.2 3.3.2.1	Visual
3.	Preservative application not as specified (Level A)	3.3.1.3	Visual MIL-P-116
4.	Preservative coated surfaces not wrapped as specified (Level A)	3.3.1.4.1	Visual
5.	Periscope not wrapped as specified (Level A, C)	3.3.1.4.2	Visual
6.	Water-vaporproof bag damaged, improperly sealed (Level A)	3.3.1.4.3	Visual
7.	Insufficient vacuum within water-vaporproof bag (Level A)	3.3.1.4.3	Visual-Tactile
8.	Periscope not cushioned and immobilized in fiberboard box as specified (Level A, C)	3.3.1.4.4 3.3.1.4.5	Visual-Tactile
9.	Unit packs not immobilized in shipping container (Level A, B, C)	3.4.1 3.4.2 3.4.3	Visual-Tactile

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TABLE I - CLASSIFICATION OF DEFECTS (Cont'd)

MAJOR: AQL 1.5		Req't Par.	Test Method
10.	Excessive gross weight (Level C)	3.4.3.1	Standard Measuring Equipment (SME)
11.	Markings illegible, incorrect, not conforming to specifications (Level A, B, C)	3.5	Visual MIL-STD-129
MINOR: AQL 4.0		**Req't Par.**	**Test Method**
41.	Shipping container not conforming to specifications (Level A, B, C)	3.4.1 3.4.2 3.4.3	SME PPP-B-585 PPP-B-601 PPP-B-621 PPP-B-636
42.	Strapping improperly applied, missing (Level A, B, C)	3.4.4	Visual PPP-B-585 PPP-B-601 PPP-B-621 PPP-B-636
43.	Workmanship (Level A, B, C)	3.7	Visual MIL-P-116

TABLE II - CLASSIFICATION OF DEFECTS

For Normal Inspection use Level L7, and Sampling Plan Table IV-A of Standard MIL-STD-105
For Reduced Inspection use Level L5, and Sampling Plan Table IV-A of Standard MIL-STD-105
AQL values are expressed in percent defective

MAJOR: AQL 4.0		Req't Par.	Test Method
1.	Heat seal (Level A)	3.3.1.4.3	MIL-P-116
2.	Quick leak (Level A)	3.3.1.4.3	MIL-P-116
3.	Preservative compound application (Level A)	3.3.1.3	MIL-P-116

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4.5 Test methods and procedures.

4.5.1 Preproduction pack. - A preproduction pack shall be inspected by the contractor or his agent and witnessed by the Government inspector to determine conformance with the requirements of this specification for the packaging and packing levels specified. The rough handling test shall be performed on the preproduction pack.

4.5.1.1 Preproduction pack failure. - Should the preproduction pack fail to pass the test for Government acceptance, production of those components causing rejection will cease until the defective characteristics have been corrected in the process used to produce the preproduction pack. When corrections in the process have been made, a new preproduction pack shall be completely inspected by the contractor and witnessed by the Government inspector.

4.5.2 Sample packs for heat seal, quick leak and preservation retention shall be selected from acceptable packs of the sample that has met the minimum inspection requirements listed in table 1.

4.5.3 Cleanliness, rough handling, heat seal, quick leak, and preservative retention test shall be conducted as specified in Specification MIL-P-116.

5. PREPARATION FOR DELIVERY

5.1 There are no applicable requirements.

6. NOTES

6.1 Ordering data. - Procurement documents should specify the following:

(a) Title, number and date of specification.
(b) Selection of applicable levels of preservation, packaging and packing.

6.2 Size and weight of packed shipping container. - The following inside dimensions, overall cubage and gross weight are approximate for two unit packaged periscopes packed in a nailed wood box for Level A:

Inside dimensions	- 36-1/2 inches long by 26-3/4 inches wide by 18-1/2 inches deep
Cubage	- 13. cubic feet
Gross weight	- 200. pounds

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6.3 Definitions.

6.3.1 Standard measuring equipment (SME). - Standard measuring equipment (SME) includes the common hand-type measuring devices which are usually stocked by commercial supply houses for ready supply (shelf items), and which are normally used by an inspector to perform dimensional inspection of items under procurement. This catagory also includes commercial testing equipment such as meters, optical comparators, etc.

6.3.2 Tactile. - Skillful manipulation or sensing.

6.4 Change of model number. - If the "XM" model number is changed to an "M" number, the title and all references to the "XM" number in this specification shall be changed to the new "M" number, pending revision of the specification.

NOTICE. When Government drawings, specifications, or other data are used for any purpose other than in connection with a definitely related Government procurement operation, the United States Government thereby incurs no responsibility nor any obligation whatsoever; and the fact that the Government may have formulated, furnished, or in any way supplied the said drawings, specifications, or other data is not to be regarded by implication or otherwise as in any manner licensing the holder or any other person or corporation, or conveying any rights or permission to manufacture, use or sell any invention that may in any way be related thereto.

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